

Minimal Lawful Implementation Spec for No-Refolding and Organizer Persistence

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Purpose

This one-page note freezes the next lawful implementation target for the HSF mesoscape program. It is not a derivation of the final no-refolding law. It is the implementation discipline to be used *before* any further churn-law tuning.

Frozen commitments

The next implementation shall adopt the following commitments:

1. No-refolding is treated as dynamical protection of committed collective relief structure.
2. Organizer identity is **function-first**, not graph-first.
3. The minimal no-refolding witness is the three-sector bundle

$$W_{\text{NR}}^{\text{min}}(M) = (W_{\text{func}}(M), W_{\text{book}}(M), W_{\text{slack}}(M)).$$

4. Sector-specific bookkeeping terms are added only when the sector is physically relevant.
5. The churn law remains frozen while bookkeeping is rewritten in named witness terms.

Minimal lawful witness

The next script shall evaluate candidate local moves using the layered witness

$$W_{\text{NR}}(M) = \alpha_{\text{func}} W_{\text{func}}(M) + \alpha_{\text{book}} W_{\text{book}}(M) + \alpha_{\text{slack}} W_{\text{slack}}(M), \quad \alpha_i \geq 0.$$

Interpretation:

- W_{func} : loss of collective relief function,
- W_{book} : loss of committed interface/bookkeeping continuity,
- W_{slack} : creation of new underused or weakly justified support directions.

Sector meaning

1. Function sector

$$W_{\text{func}}(M) = 1 - P_M(M)$$

for a function-preservation score $P_M \in [0, 1]$ built from admissible temporary MR-family proxies.

Official temporary proxies:

- local MI bundle,
- correlator profile,
- retained committed-interface activity,
- birth-justification / parent-relief witness when evaluating births.

2. Bookkeeping sector

$$W_{\text{book}}(M) = 1 - P_K(M)$$

for a bookkeeping-preservation score $P_K \in [0, 1]$ built from KR-family proxies.

Official temporary proxies:

- endpoint/interface bookkeeping continuity,
- committed-interface activity continuity,
- sector-specific bookkeeping continuity when explicitly represented.

3. Slack sector

$$W_{\text{slack}}(M) = \Delta S_+(M)$$

for the positive increase in underused or weakly justified support/slack burden.

Official temporary proxies:

- active-channel rank loss / slack increase,
- influence-rank profile degradation,
- influence spread inflation when used only as a secondary slack diagnostic,
- support padding without corresponding preserved function.

Lawful move classifier

A candidate move M in local region R shall be classified as follows.

Lawful re-expression

A move counts as lawful re-expression only if it satisfies all three conditions:

$$\begin{aligned} W_{\text{func}}(M) &\leq \varepsilon_f, \\ W_{\text{book}}(M) &\leq \varepsilon_k, \\ W_{\text{slack}}(M) &\leq \varepsilon_s. \end{aligned}$$

Interpretation:

- a) the same collective relief function remains sufficiently present,
- b) enough bookkeeping continuity remains for physical identity,
- c) support has not been preserved merely by slack-heavy padding.

Unlawful refolding

A move counts as refolding-like if one or more of the following hold:

- a) substantial function loss,
- b) major bookkeeping discontinuity,
- c) large slack creation,
- d) sector-specific bookkeeping failure when that sector is active.

Audit / unresolved

If the three sectors disagree strongly or the local evidence is insufficient, the move must be logged as unresolved rather than silently accepted by graph convenience.

Sector-specific add-ons

The general witness must remain general. Extra bookkeeping terms are added only when relevant:

$$W_{\text{NR}}^{\text{sector}}(M) = W_{\text{NR}}(M) + \lambda_{\text{sec}} W_{\text{sec}}(M).$$

For the next implementation:

- Jordan–Wigner/string preservation is a **fermionic add-on**, not the whole general meaning of no-refolding.
- Gauge/interface compatibility is a **gauge-sector add-on**.
- Collective response / mode preservation is a **mesoscopic object add-on**.

Calibration rule

Every scalable implementation of the function-bearing sector must be calibrated against exact small-region or small- N cases whenever feasible.

The calibration must answer:

1. Does the scalable witness preserve representative move rankings?
2. Does it distinguish lawful re-expression from real function loss?
3. Does it distinguish justified birth from redundant support cloning?

If the answer is no, the proxy is not yet acceptable.

Logging requirement

For every candidate move, the script must log at minimum:

1. move type,
2. local region identifier,
3. $W_{\text{func}}(M)$,
4. $W_{\text{book}}(M)$,
5. $W_{\text{slack}}(M)$,
6. any active sector-specific witness,
7. final classifier: lawful / refolding-like / unresolved.

Combined scores may be logged, but the named sector scores must not be hidden.

What is forbidden

The next implementation must **not** do any of the following:

1. replace W_{NR} with shell size, edge count, triangle count, support count, or graph persistence alone,
2. treat dense reduced-state identity as the default general organizer criterion,
3. use influence spread as the primary lawful term,
4. present temporary proxies as fully derived quantities,
5. present chosen weights as first-principles constants,
6. claim that no-refolding has already been fully derived.

Retirement rule

Support retirement is lawful only under post-redistribution conditions:

1. the retiring support is no longer carrying essential organizer function,
2. neighboring support now carries the relevant function,
3. bookkeeping continuity remains acceptable,
4. the organizer either survives or is genuinely dissolved rather than silently forgotten.

One-sentence operational summary

For the next lawful implementation, no-refolding is measured by a minimal three-sector witness tracking preserved relief function, preserved bookkeeping continuity, and suppression of slack-heavy fake preservation, with sector-specific bookkeeping added only when physically relevant.